

and taking a supremum over y we obtain the bound:

$$\sup_{y \in \mathcal{X}} \left| \frac{d}{d\varepsilon} \mathcal{D}(\theta; \mathbb{P}_{\varepsilon, y}) \right| \leq 2 \sup_{y \in \mathcal{X}} |d(\theta, y)| \leq 2\gamma(\theta),$$

where the last inequality holds since γ fulfil condition 1. Using this bound, we check the two conditions of Matsubara et al. (2022b).

1. $\sup_{\theta \in \Theta} \sup_{y \in \mathcal{X}} \left| \frac{d}{d\varepsilon} \mathcal{D}(\theta; \mathbb{P}_{\varepsilon, y}) \right| \pi(\theta) \leq \sup_{\theta \in \Theta} 2\gamma(\theta) \pi(\theta) < \infty$, and
2. $\int_{\Theta} \sup_{y \in \mathcal{X}} \left| \frac{d}{d\varepsilon} \mathcal{D}(\theta; \mathbb{P}_{\varepsilon, y}) \right| \pi(\theta) d\theta \leq \int_{\Theta} 2\gamma(\theta) \pi(\theta) d\theta < \infty$,

where the last inequalities hold because γ meets conditions 2 and 3. Therefore, by virtue of Matsubara et al. (2022b) the posterior is globally bias-robust. ■

B.4. Proof of Proposition 3.2

In this Subsection we provide the proof of Proposition 3.2. The strategy of the proof is simple: We show that $d = d_m$ admits a natural function γ that satisfies the conditions of Proposition B.1.

Proof. From Proposition B.1, it is sufficient to find a function γ such that:

$$\sup_{y \in \mathcal{X}} \left| \underbrace{\|m^\top(y) \nabla \log p_\theta(y)\|_2^2}_{(1)} + 2 \underbrace{\nabla \cdot m(y) m^\top(y) \nabla \log p_\theta(y)}_{(2)} \right| \leq \gamma(\theta).$$

Now, following from the form of m in Proposition 3.2 and the fact that p_θ is an exponential family member as in (5), we have:

$$(1) = \|m^\top(y) \nabla \log p_\theta(y)\|_2^2 = \sum_{i=1}^d (m^\top(y) \nabla \log p_\theta(y))_i^2 = \sum_{i=1}^d \frac{(\nabla r(x)\theta)_i^2}{1 + (\nabla r(x)\theta^*)^2_i} \leq \sum_{i=1}^d \frac{(\nabla r(x)\theta)_i^2}{(\nabla r(x)\theta^*)^2_i}.$$

Using the fact that $\|x\|_2^2 \leq \|x\|_1^2 \leq d\|x\|_2^2$ for $x \in \mathbb{R}^d$, we have:

$$\sum_{i=1}^d \frac{(\nabla r(x)\theta)_i^2}{(\nabla r(x)\theta^*)^2_i} \leq \sum_{i=1}^d \frac{p\|\theta\|_2^2}{\|\theta^*\|_2^2} = \frac{dp\|\theta\|_2^2}{\|\theta^*\|_2^2} =: \gamma_1(\theta).$$

For the second expression, we have:

$$\begin{aligned} (2) &= |\nabla \cdot m(y) m^\top(y) \nabla \log p_\theta(y)| = \left| \sum_{i=1}^d \frac{\partial}{\partial x_i} (m(y) m^\top(y) \nabla \log p_\theta(y))_i \right| \\ &= \left| \sum_{i=1}^d \frac{\partial}{\partial x_i} \left(\frac{(\nabla r(x)\theta)_i}{1 + (\nabla r(x)\theta^*)^2_i} \right) \right| \\ &= \left| \sum_{i=1}^d \frac{(\nabla^2 r(x)\theta)_{ii} (1 + (\nabla r(x)\theta^*)^2_i) - 2(\nabla r(x)\theta^*)_i (\nabla r(x)\theta)_i (\nabla^2 r(x)\theta^*)_{ii}}{(1 + (\nabla r(x)\theta^*)^2_i)^2} \right| \\ &\leq \sum_{i=1}^d \left| \frac{(\nabla^2 r(x)\theta)_{ii}}{1 + (\nabla r(x)\theta^*)^2_i} \right| + 2 \left| \frac{(\nabla r(x)\theta^*)_i (\nabla r(x)\theta)_i (\nabla^2 r(x)\theta^*)_{ii}}{(1 + (\nabla r(x)\theta^*)^2_i)^2} \right|. \end{aligned}$$

For most distributions of interest, including Gaussians, exponentials, (inverse) Gamma, and Beta distributions, $\left| \frac{(\nabla^2 r(x)\theta)_{ii}}{1 + (\nabla r(x)\theta^*)^2_i} \right|$ is bounded for every $\theta \in \Theta$, then:

$$\begin{aligned} \sum_{i=1}^d \left| \frac{(\nabla^2 r(x)\theta)_{ii}}{1 + (\nabla r(x)\theta^*)^2_i} \right| + 2 \left| \frac{(\nabla r(x)\theta^*)_i (\nabla r(x)\theta)_i (\nabla^2 r(x)\theta^*)_{ii}}{(1 + (\nabla r(x)\theta^*)^2_i)^2} \right| &\leq dC(\theta) + 2C(\theta) \sum_{i=1}^d \left| \frac{(\nabla r(x)\theta^*)_i (\nabla r(x)\theta)_i}{(1 + (\nabla r(x)\theta^*)^2_i)^2} \right| \\ &\leq dC(\theta) (1 + 2d \frac{\|\theta\|_2^2}{\|\theta^*\|_2^2}) =: \gamma_2(\theta). \end{aligned}$$

Defining $\gamma(\theta) := \gamma_1(\theta) + \gamma_2(\theta)$ we have :

$$\sup_{y \in \mathcal{X}} \left| \|m^\top(y) \nabla \log p_\theta(y)\|_2^2 + 2 \nabla \cdot m(y) m^\top(y) \nabla \log p_\theta(y) \right| \leq \gamma(\theta).$$

Now we are in a position to verify conditions (I) and (II) of Proposition B.1. Since $\gamma(\theta)$ is a polynomial function, $\pi(\theta)$ is a squared exponential prior, and the squared exponential has infinitely many moments, it is clear that:

$$\begin{aligned} \sup_{\theta \in \Theta} \pi(\theta) \gamma(\theta) &< \infty, \\ \int_{\Theta} \pi(\theta) \gamma(\theta) d\theta &< \infty, \end{aligned}$$

which completes the proof. ■

B.5. Boundary and smoothness conditions

In this subsection, we review the boundary and smoothness conditions discussed in Section 3.1, alongside the DSM extension to more general domains $\mathcal{X}$.

The smoothness conditions needed to get the expansion of $\mathcal{D}_m$ as in Equation (3) for $\mathcal{X} = \mathbb{R}^d$ are the following:

Lemma B.2. *If p_θ is twice-differentiable, and $p_0 m m^\top \nabla \log p_\theta, \nabla \cdot (p_0 m m^\top \nabla \log p_\theta) \in L^1(\mathbb{R}^d)$, then we can rewrite $\mathcal{D}_m$ as in Equation (3).*

Yu et al. (2019) extended the DSM to densities with non-negative support, i.e. $\mathcal{X} = \mathbb{R}_{\geq 0}^d$.

Theorem B.3. *Suppose that $\log p_0(x)$ and m are continuously differentiable almost everywhere on $\mathbb{R}_+^d$ and $\log p_\theta$ is twice continuously differentiable with respect to x on $\mathbb{R}_+^d$. Furthermore, we assume the boundary condition,*

$$\lim_{|x^{(i)}| \rightarrow \infty} p_0(x) m_{ii}^2(x) \partial_i \log p_\theta(x) - \lim_{|x^{(i)}| \rightarrow 0^+} p_0(x) m_{ii}^2(x) \partial_i \log p_\theta(x) = 0, \forall i \in \{1, \dots, d\},$$

where x_i is the i -dimension of x . Then, we can rewrite $\mathcal{D}_m$ as in Equation (3)

Later, Liu et al. (2022) extended the DSM to densities with support in a Lipschitz Domain, which, intuitively speaking, are bounded connected open domains whose local boundary is a level set of some Lipschitz function.

Theorem B.4. *Assume $\mathcal{X} \subset \mathbb{R}^d$ is a Lipschitz domain. Suppose $p_0, \partial_i \log p_\theta \in H^1(\mathcal{X})$ and that for any $z \in \partial\mathcal{X}$ it holds that*

$$\lim_{x \rightarrow z} p_0(x) m_{ii}^2(x) \partial_i \log p_\theta(x) v_i(z) = 0, \forall i \in \{1, \dots, d\},$$

where $x \rightarrow z$ takes any point sequence converging to $z \in \partial\mathcal{X}$ into account, $v = (v_1, \dots, v_d)$ is the unit outward normal vector on $\partial\mathcal{X}$, and $H^1(\mathcal{X})$ is the Sobolev-Hilbert space. Then, we can rewrite $\mathcal{D}_m$ as in Equation (3).

The Sobolev-Hilbert space is defined as follows:

$$H^1(\mathcal{X}) = \{f \in L^2(\mathcal{X}) \mid \|f\|_{L^2(\mathcal{X})}^2 + \sum_{i=1}^d \|D_i f\|_{L^2(\mathcal{X})}^2 < \infty\},$$

where D_i is the weak derivative corresponding to ∂_i and $\|f\|_{L^2(\mathcal{X})} = \sqrt{\int_{\mathcal{X}} |f(x)|^2 dx}$.

Gamma distribution Let p_θ be a gamma distribution, and $\mathcal{X} = \mathbb{R}_+$. Assume p_0 bounded from above and that $\log p_0(x)$ is continuously differentiable almost everywhere on $\mathbb{R}_+$. Then for m as in Proposition 3.2, we have

$$\begin{aligned} \lim_{|x| \rightarrow 0^+} p_0(x) m^2(x) \partial \log p_\theta(x) &= \lim_{|x| \rightarrow 0^+} p_0(x) \frac{\nabla r(x) \theta + \nabla b(x)}{1 + (\nabla r(x) \theta^*)^2_i} \\ &= \lim_{|x| \rightarrow 0^+} p_0(x) \frac{\frac{\theta_1}{x} - \theta_2}{1 + (\frac{\theta_1}{x} - \theta_2^*)^2} \\ &= \lim_{|x| \rightarrow 0^+} p_0(x) \frac{x \theta_1 - x^2 \theta_2}{x + (\theta_1^* - x \theta_2^*)^2} \\ &= 0. \end{aligned}$$

The last equality holds since p_0 is bounded. Now, for the second boundary condition:

$$\lim_{|x| \rightarrow \infty} p_0(x) m^2(x) \partial \log p_\theta(x) = \lim_{|x| \rightarrow \infty} p_0(x) \frac{\nabla r(x) \theta + \nabla b(x)}{1 + (\nabla r(x) \theta^*)^2_i} = \lim_{|x| \rightarrow \infty} p_0(x) \frac{\frac{\theta_1}{x} - \theta_2}{1 + (\frac{\theta_1}{x} - \theta_2^*)^2} = 0.$$

The last equality holds since p_0 a density, therefore, $\lim_{|x| \rightarrow \infty} p_0(x) = 0$. Then the m proposed in Proposition 3.2 satisfies the boundary conditions in Theorem B.3 for the gamma distribution.

Exponential distribution Let p_θ be an exponential distribution, and $\mathcal{X} = \mathbb{R}_+$. Assume p_0 bounded, such that $\log p_0(x)$ is continuously differentiable almost everywhere on $\mathbb{R}_+$. Then for m such as in Proposition 3.2 :

$$\lim_{|x| \rightarrow 0^+} p_0(x) m^2(x) \partial \log p_\theta(x) = \lim_{|x| \rightarrow 0^+} p_0(x) \frac{\nabla r(x) \theta + \nabla b(x)}{1 + (\nabla r(x) \theta^*)^2_i} = \lim_{|x| \rightarrow 0^+} p_0(x) = \frac{\theta}{1 + \theta^2} \lim_{|x| \rightarrow 0^+} p_0(x).$$

The term for the second limit would be similar:

$$\lim_{|x| \rightarrow \infty} p_0(x) m^2(x) \partial_i \log p_\theta(x) = \frac{\theta}{1+\theta^2} \lim_{|x| \rightarrow \infty} p_0(x).$$

Therefore, the expression in Theorem B.3 looks as follows:

$$\lim_{|x^{(i)}| \rightarrow \infty} p_0(x) m_{ii}^2(x) \partial_i \log p_\theta(x) - \lim_{|x^{(i)}| \rightarrow 0+} p_0(x) m_{ii}^2(x) \partial_i \log p_\theta(x) = \frac{\theta}{1+\theta^2} (\lim_{|x| \rightarrow \infty} p_0(x) - \lim_{|x| \rightarrow 0+} p_0(x)).$$

Then the m proposed in Proposition 3.2 satisfies the boundary conditions in Theorem B.3 if

$$\lim_{|x| \rightarrow \infty} p_0(x) = \lim_{|x| \rightarrow 0+} p_0(x).$$

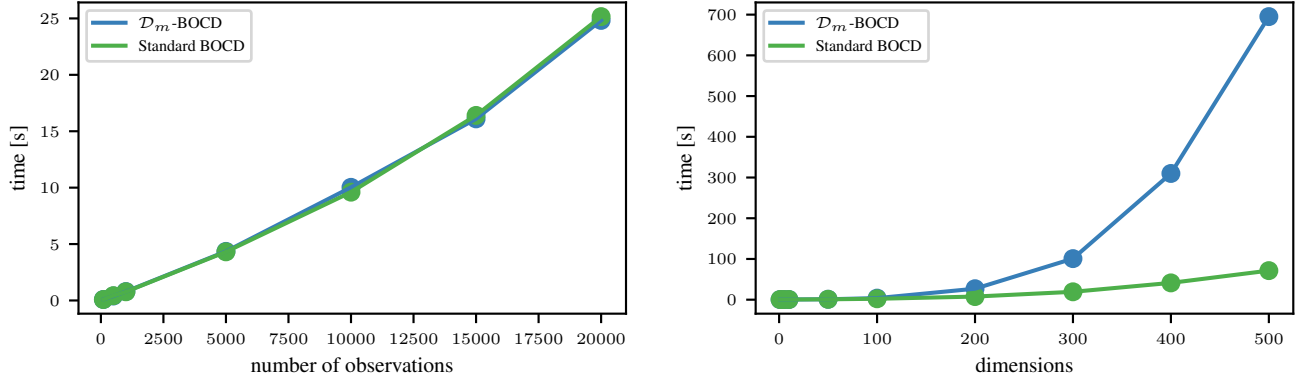
C. Additional Details on Numerical Experiments

In this section we give additional details on the numerical experiments of Section 4. We provide the exact prior and ω used in each experiment. Moreover, we present an extra numerical experiment to compare the computational complexity of standard BOCD and $\mathcal{D}_m$ -BOCD in Appendix C.1.

C.1. Computational complexity

The computational complexity of both standard BOCD and $\mathcal{D}_m$ -BOCD is linear in the number of data points. To verify that this theoretical complexity mirrors the practical computational overhead, we generate samples from a Gaussian distribution with 1 CP where the mean varies. We vary the sample size from $T = 100$ up to $T = 20000$. We fit a Gaussian distribution with the correct variance taken as fixed in both the $\mathcal{D}_m$ -BOCD and the standard BOCD. As shown in Figure 8a, both methods are equally fast for any number of observations.

Although the computational complexity of $\mathcal{D}_m$ -BOCD is linear in the data, it is quadratic in the dimension of the observations, in particular, is $\mathcal{O}(T(p^2 + d^2))$. To observe this in practice, we consider the same settings as before, but now we fix the sample size to $T = 100$ and vary the data dimensions from $d = 1$ up to $d = 500$. Figure 8b shows that both methods take practically the same time when d is less than 100.



(a) Overall time in seconds versus number of observations. We observe that both methods are equally fast for any number of observations.

(b) Overall time in seconds versus the dimension of the observations. We observe that both methods are practically equally fast when the dimension of the observations is less than 100.

Figure 8. Comparison between $\mathcal{D}_m$ -BOCD and the standard BOCD. blue line for $\mathcal{D}_m$ -BOCD, and green line for standard BOCD.

The previous setting is where $\mathcal{D}_m$ -BOCD clearly shows its advantage over the β -BOCD due to its completely closed form, making it nearly equivalent, in terms of computational overhead, to standard BOCD. In more complex settings, the posterior predictive may not be available in closed form; hence, we approximate it by sampling from $\pi_\omega^{\mathcal{D}_m}$. It might not be obvious how this is a significant advantage over the β -BOCD framework. To demonstrate that $\mathcal{D}_m$ -BOCD is faster than β -BOCD, we generate samples from a Gaussian distribution with 1 CP, where the mean and the variance change. We vary the sample size from $T = 100$ to $T = 20000$ and fit a Gaussian distribution in the $\mathcal{D}_m$ -BOCD, β -BOCD and the standard BOCD. In Figure 9, we observe that although the standard BOCD is faster than $\mathcal{D}_m$ -BOCD, our method is considerably faster than the β -BOCD for any number of observations.